

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			award (1) for either of following - prevents acid (-spray) escaping <u>only</u> allows carbon dioxide to escape			1	1		1
	(b)			take the mass (after 10 s) away from 107.75 / the original mass			1	1		1
	(c)			0.0082 / 8.2×10^{-3} (3) must be to 2 sig figs if incorrect award (1) each for either of following appropriate change in mass e.g. 0.44 – 0.30 / 0.14 corresponding change in time e.g. 30 – 13 / 17 ecf possible from error in graph reading but must have calculated a change in mass and change in time		3		3	3	
	(d)	(i)		award (2) for all points plotted correctly – tolerance $\pm\frac{1}{2}$ square award (1) for any four points plotted correctly award (1) for (smooth) curve drawn through points ecf possible from incorrectly plotted points		3		3	3	
		(ii)		acid has lower concentration (1) so fewer particles present (in the same volume) (1) lower chance of successful collisions / lower frequency of successful collisions / fewer successful collisions (per second) (1)	2		1	3		

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	(e)			curve to the left of and above graph A starting from (0,0) and to a maximum height of 0.88 – tolerance $\pm\frac{1}{2}$ square			1	1		
				Question 7 total	2	6	4	12	6	2